MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 5)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. In the following figure, P and Q are points of trisection of line segment AB:

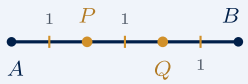
$$A \text{ ——— } P \text{ ——— } Q \text{ ——— } B$$

the value of $\frac{AB}{PB} =$ [1]

(A) 1 **(B)** 1.5 **(C)** $\frac{2}{3}$ **(D)** 2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 1.5



Given: Points P and Q trisect segment AB into three equal parts, so $AP = PQ = QB = k$. Hence $AB = 3k$ and $PB = 2k$.

We know the trisection property:

Formula Used: Trisection property: $AP = PQ = QB = k \implies AB = 3k,\ PB = PQ + QB = 2k$

Therefore:

$$\frac{AB}{PB} = \frac{3k}{2k} = \frac{3}{2} = \boxed{1.5}$$

Q2. If the points $A(1,2)$, $O(0,0)$ and $C(a,b)$ are collinear, then: [1]

(A) $a=b$ **(B)** $a=2b$ **(C)** $2a=b$ **(D)** $a=-b$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (C) $2a=b$

Given: $A(1,2)$, $O(0,0)$, and $C(a,b)$ are collinear.

Formula Used: Collinear points lie on the same line: coordinate ratios scale proportionally from O(0,0).

If $a=0$, the line through $(0,0)$ and $(1,2)$ forces $C=(0,0)$, giving $b=0$ ($2a=b$ holds). For $a\neq 0$, suppose $O(0,0)$ divides AC in ratio $k:1$ (using the section formula $\frac{m_1x_2+m_2x_1}{m_1+m_2}=0$ and $\frac{m_1y_2+m_2y_1}{m_1+m_2}=0$):

$$ka+1=0 \implies k=-\frac{1}{a}, \qquad \text{and} \qquad kb+2=0 \implies -\frac{b}{a}+2=0 \implies \frac{b}{a}=2 \implies \boxed{\mathbf{2a=b}}$$

Q3. **Assertion (A):** The point $(-2,4)$ divides the line segment joining the points $(-4,8)$ and $(5,-10)$ in the ratio $2:7$ internally.

Reason (R): If three points P, Q and R are collinear, then $PQ+QR=PR$. [1]

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B)

Evaluating Assertion (A): Check the internal section formula for $A(-4,8)$ and $B(5,-10)$ in ratio $2:7$:

Formula Used: $(x,y)=\left(\frac{m_1x_2+m_2x_1}{m_1+m_2},\ \frac{m_1y_2+m_2y_1}{m_1+m_2}\right)$

$$x=\frac{2(5)+7(-4)}{2+7}=\frac{10-28}{9}=\frac{-18}{9}=-2, \qquad y=\frac{2(-10)+7(8)}{2+7}=\frac{-20+56}{9}=\frac{36}{9}=4$$

Since $(-2,4)$ matches exactly, Assertion (A) is **true**.

Evaluating Reason (R): When three collinear points P,Q,R lie on segment PR in order with Q between P and R, $PQ+QR=PR$. Hence Reason (R) is **true** as a standard collinearity statement. However, Assertion (A) is established by the **section formula**, not by the segment-addition rule in (R). Thus, both (A) and (R) are true, but (R) is not the correct explanation of (A).

Q4. The coordinates of the midpoint of the line segment joining the points $(-5, 7)$ and $(-1, 3)$ are: [1]

(A) $(-3, 5)$ **(B)** $(3, 5)$ **(C)** $(-3, -5)$ **(D)** $(5, -3)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** $(-3, 5)$

Given: Points $(-5, 7)$ and $(-1, 3)$.

Formula Used: Midpoint $M = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$

$$M = \left(\frac{-5 + (-1)}{2}, \frac{7 + 3}{2}\right) = \left(\frac{-6}{2}, \frac{10}{2}\right) = \boxed{(-3, 5)}$$

Q5. If $\left(\dfrac{a}{3}, 4\right)$ is the midpoint of the line segment joining the points $Q(-6, 5)$ and $R(-2, 3)$, then the value of a is: [1]

(A) -4 **(B)** -12 **(C)** 12 **(D)** -6

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B)** -12

Given: $Q(-6, 5)$, $R(-2, 3)$, midpoint $P\left(\frac{a}{3}, 4\right)$.

Formula Used: $x_{\text{mid}} = \dfrac{x_1 + x_2}{2}$

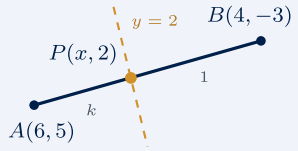
$$\frac{a}{3} = \frac{-6 + (-2)}{2} = \frac{-8}{2} = -4 \implies a = -4 \times 3 = \boxed{-12}$$

SECTION B: Short Answer Type Questions

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Q6. Calculate the ratio in which the line segment joining the points $A(6, 5)$ and $B(4, -3)$ is divided by the line $y = 2$. Also, find the point of intersection. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Finding Ratio Using Ordinate Condition $y = 2$ [1.5 Marks]

Given: Endpoints $A(6, 5)$ and $B(4, -3)$. Since the dividing point lies on the line $y = 2$, let $P(x, 2)$ divide segment AB internally in the ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\dfrac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \dfrac{m_1 y_2 + m_2 y_1}{m_1 + m_2}\right)$

Equating the y-coordinate of P to 2:

$$\frac{k(-3) + 1(5)}{k + 1} = 2 \implies -3k + 5 = 2(k + 1) \implies -3k + 5 = 2k + 2 \implies 5k = 3 \implies k = \frac{3}{5}$$

Hence, the ratio is **3 : 5**.

Step 2: Finding Coordinates of P [1.5 Marks]

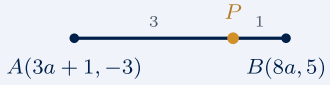
Using $m_1 = 3$, $m_2 = 5$, $x_1 = 6$, and $x_2 = 4$:

$$x = \frac{3(4) + 5(6)}{3 + 5} = \frac{12 + 30}{8} = \frac{42}{8} = \frac{21}{4}$$

$$\boxed{\text{Ratio } = \mathbf{3 : 5} \quad \text{and} \quad P = \left(\mathbf{\frac{21}{4}}, \mathbf{2}\right)}$$

Q7. If $P(9a-2,\ -b)$ divides the line segment joining $A(3a+1,\ -3)$ and $B(8a,\ 5)$ in the ratio $3:1$, find the values of a and b. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Applying Section Formula for $3:1$ [1 Mark]
Given: Points $A(3a+1,-3)$ and $B(8a,5)$, dividing point $P(9a-2,-b)$, and ratio $m_1:m_2=3:1$.

We know the internal section formula:

Formula Used: $P(x,y)=\left(\dfrac{m_1x_2+m_2x_1}{m_1+m_2},\ \dfrac{m_1y_2+m_2y_1}{m_1+m_2}\right)$

Step 2: Equating Abscissa and Ordinate to Solve for a, b [2 Marks]
For the x-coordinate:

$$9a-2=\frac{3(8a)+1(3a+1)}{3+1}=\frac{24a+3a+1}{4}=\frac{27a+1}{4}$$

Multiplying both sides by 4:

$$4(9a-2)=27a+1 \implies 36a-8=27a+1 \implies 36a-27a=1+8 \implies 9a=9 \implies \mathbf{a=1}$$

For the y-coordinate:

$$-b=\frac{3(5)+1(-3)}{3+1}=\frac{15-3}{4}=\frac{12}{4}=3 \implies \mathbf{b=-3}$$

$$\boxed{\mathbf{a=1} \quad \text{and} \quad \mathbf{b=-3}}$$

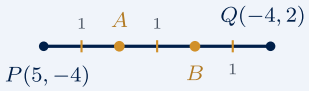
SECTION C: Long Answer & Competency Based Questions MATHS MANTHAN | (9 Marks)

Q8. Find the coordinates of the points of trisection of the line segment joining the points $P(5,-4)$ and $Q(-4,2)$. **[5]**

— OR —

Find the ratio in which the point $P(6,k)$ divides the line segment joining the points $M(4,2)$ and $N(8,-4)$. Hence find the value of k. **[5]**

▶ **SOLUTION (OPTION I):** [Total: 5 Marks]



Step 1: Section Ratios for Trisection Points A and B [1 Mark]
Let A and B trisect PQ so that $PA=AB=BQ$. Thus, point A divides PQ internally in ratio $1:2$, and point B divides PQ internally in ratio $2:1$.

We know the internal section formula:

Formula Used: $(x,y)=\left(\dfrac{m_1x_2+m_2x_1}{m_1+m_2},\ \dfrac{m_1y_2+m_2y_1}{m_1+m_2}\right)$

Step 2: Finding Point A (Ratio $1:2$) [2 Marks]

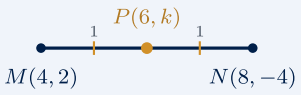
$$A=\left(\frac{1(-4)+2(5)}{1+2},\ \frac{1(2)+2(-4)}{1+2}\right)=\left(\frac{-4+10}{3},\ \frac{2-8}{3}\right)=\left(\frac{6}{3},\ \frac{-6}{3}\right)=(\mathbf{2},\mathbf{-2})$$

Step 3: Finding Point B (Ratio $2:1$) [2 Marks]

$$B=\left(\frac{2(-4)+1(5)}{2+1},\ \frac{2(2)+1(-4)}{2+1}\right)=\left(\frac{-8+5}{3},\ \frac{4-4}{3}\right)=\left(\frac{-3}{3},\ \frac{0}{3}\right)=(\mathbf{-1},\mathbf{0})$$

$$\boxed{\text{Points of trisection } =(\mathbf{2},\mathbf{-2}) \text{ and } (\mathbf{-1},\mathbf{0})}$$

► **SOLUTION (OPTION II):** [OR: 5 Marks]



Step 1: Finding the Internal Ratio [3 Marks]

Given: Endpoints $M(4,2)$, $N(8,-4)$, and point $P(6,k)$. Let P divide segment MN internally in the ratio $r:1$.

We know the internal section formula:

Formula Used: $P(x,y) = \left(\dfrac{m_1x_2+m_2x_1}{m_1+m_2}, \dfrac{m_1y_2+m_2y_1}{m_1+m_2}\right)$

Equating the x-coordinate of P to 6:

$$\frac{r(8)+1(4)}{r+1} = 6 \implies 8r+4 = 6r+6 \implies 2r = 2 \implies r = 1 \implies \text{Ratio} = \mathbf{1:1}$$

Step 2: Finding the Value of k [2 Marks]

Since ratio is $1:1$ (P is the midpoint of MN), substituting $r=1$ into the y-coordinate:

$$k = \frac{1(-4)+1(2)}{1+1} = \frac{-2}{2} = \mathbf{-1}$$

Ratio $= \mathbf{1:1}$ and $\mathbf{k = -1}$

Q9. Competency Case Study: A municipal corporation maps three community utility centers situated at points $P(2,3)$, $Q(6,7)$ and $R(8,9)$ on a city plan. [4]

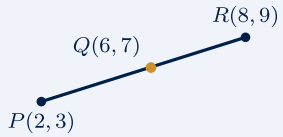
(i) Find the distance between utility center P and center Q.

(ii) Find the coordinates of a service station S located midway between P and R.

(iii) Check whether points P, Q and R lie on a straight road (are collinear) by comparing distance sums.

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▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Given: Utility centers at $P(2,3)$, $Q(6,7)$, and $R(8,9)$. We apply the distance and midpoint formulas to find PQ, service station S, and check collinearity ($PQ + QR = PR$).

We know the distance and midpoint formulas:

Formula Used:
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

(i) Distance PQ Between Centers $P(2,3)$ and $Q(6,7)$: [1 Mark]

$$PQ = \sqrt{(6-2)^2 + (7-3)^2} = \sqrt{4^2 + 4^2} = \sqrt{32} = \mathbf{4\sqrt{2}} \text{ units}$$

(ii) Midpoint Service Station S Between $P(2,3)$ and $R(8,9)$: [1 Mark]

$$S = \left(\frac{2+8}{2}, \frac{3+9}{2} \right) = \left(\frac{10}{2}, \frac{12}{2} \right) = (\mathbf{5}, \mathbf{6})$$

(iii) Checking Collinearity by Comparing Distance Sums: [2 Marks]

Computing remaining distances QR and PR:

$$QR = \sqrt{(8-6)^2 + (9-7)^2} = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2} \text{ units}$$

$$PR = \sqrt{(8-2)^2 + (9-3)^2} = \sqrt{6^2 + 6^2} = \sqrt{72} = 6\sqrt{2} \text{ units}$$

Comparing $PQ + QR$ with PR:

$$PQ + QR = 4\sqrt{2} + 2\sqrt{2} = 6\sqrt{2} = PR$$

$$\boxed{PQ = 4\sqrt{2} \text{ units}, \quad S = (5,6), \quad \text{Yes, } P, Q, R \text{ are collinear.}}$$

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